Polarized versus Threshold Intensity Distribution in Sub-Elite Middle-Distance Runners: A 16-Week Randomized Trial

Kari Ellingsen, Peter Mwangi

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Abstract

We randomized 34 sub-elite middle-distance runners to 16 weeks of polarized (about 80% low intensity, 20% high) or threshold-focused training at matched total load. Both groups improved 1500 m time-trial performance; the polarized group improved more (2.1% vs 1.4%). Velocity at lactate threshold improved similarly in both groups. Runners in the polarized group reported lower perceived fatigue across the block.

1 Introduction

Endurance coaches debate how to distribute intensity. Polarized programs keep most volume easy and add a small amount of very hard work; threshold programs spend more time near lactate threshold. Evidence in middle-distance runners, whose events are short and intense, is limited.

2 Methods

34 runners (1500 m personal bests 3:52-4:30 for men, 4:25-5:05 for women) were randomized to polarized (POL, n = 17) or threshold (THR, n = 17) training for 16 weeks. Weekly training load was matched with session RPE. Intensity zones were set from a lactate test before the block.

Variable	POL pre	POL post	THR pre	THR post
1500 m time trial (s)	262.4	256.9	263.1	259.4
vLT (km/h)	16.1	16.8	16.0	16.7
VO2max (ml/kg/min)	62.3	64.0	61.9	62.8
Weekly volume (h)	7.8	8.4	7.9	8.1

3 Results

Both groups improved time-trial performance (POL 2.1%, THR 1.4%; group x time p = 0.04). Velocity at lactate threshold improved similarly. Perceived fatigue (weekly questionnaire) was lower in POL from week 6 onward.

4 Discussion

For middle-distance runners, keeping easy days easy and concentrating hard work appears at least as effective as a threshold-heavy program and better tolerated. Base phases built on long low-intensity volume remain a reasonable foundation.

5 Limitations

Small sample, one training group, and a block too short to observe long-term adaptations.

References

[1] Zone definitions, lactate protocols and individual training logs are available in the supplementary material.